

# Senior Software Engineer

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## Nicolas Molina Monroy

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## Profile

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Senior Software Engineer with over 13 years of experience delivering scalable, user-centric products across both frontend and backend systems. I excel in leading architecture and execution, particularly in distributed systems and event-driven design, leveraging technologies such as FastAPI/Python and Node.js.

My expertise extends to integrating LLMs, RAG, and voice AI, enhancing user experiences through innovative solutions. Recognized as a Google Developer Expert and Microsoft MVP, I have a proven track record of building robust, efficient systems that align with business goals and elevate user experience. My work involves collaborating cross-functionally to ensure seamless feature delivery and system performance at scale

## Experience

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### Senior Software Engineer at dotCMS

May 2024 — Present

Remote, USA

- Implement an AI chat experience within the platform using Generative UI to build dynamic components tailored to user needs. This solution addresses common inquiries, automates form-filling, and streamlines cross-platform navigation.
- Enhance the dotCMS platform by leading the design and development of scalable features, ensuring alignment with R&D and product goals. Focus on improving performance and reliability through efficient code contributions and collaboration.
- Developed scalable and reusable components to enhance performance and maintainability.

### LangGraph for AI Agents Instructor at Platzi

Sept 2025

Remote, Colombia

- Authored and delivered the specialized "LangGraph for AI Agents" curriculum, defining best practices for building stateful, autonomous workflows. Teaches advanced patterns for orchestrating multi-agent systems and managing application state reliability.
- Instructs senior developers on integrating LLMs into production-grade backend architectures, focusing on fault tolerance, tool usage, and scalable design patterns.
- Translates complex engineering concepts—including distributed systems logic and scalable backend strategies—into actionable, high-quality code implementations for the developer community.

### Senior Full-Stack Developer at Platzi

May 2023 — Jan 2024

Remote, Colombia

- Architected and scaled "Ada," the AI Tutor, to support 6,000+ concurrent students with low latency. Built a distributed, serverless infrastructure using HonoJS on Cloudflare Workers (Edge Computing) to ensure high availability and operational stability.
- Engineered the LLM orchestration layer using LangChainJS, integrating GPT-4 and RAG (Retrieval-Augmented Generation) techniques. Optimized prompt engineering workflows to deliver precise, context-aware educational support.
- Led the backend development of the English Placement Test using Python (FastAPI). Implemented a Computerized Adaptive Testing (CAT) algorithm and the Whisper Model for accurate voice analysis and automated scoring.

### Full Stack Developer at Platzi

Jun 2020 — May 2023

Remote, Colombia

- Engineered a high-performance, browser-based code execution environment ("Coding Playground"). Leveraged Pyodide (WebAssembly) to run full Python kernels client-side, significantly reducing backend computational load while enabling secure, interactive challenges for Python, SQL, and JavaScript.
- Designed and maintained the "Platzi Fake Store API," a public, high-availability RESTful service used by the developer community for e-commerce prototyping. Ensured strict API contract reliability and scalable data delivery.
- Spearheaded the backend architecture for the Platzi Master Platform using NestJS and PostgreSQL. Designed complex relational database schemas to capture high-volume analytics on student-mentor interactions, ensuring data integrity and query performance.
- Orchestrated CI/CD pipelines using Docker and GitHub Actions, automating deployment workflows and ensuring environment consistency across development and production.

### Senior Developer at Lulla Care

May 2019 — Jun 2020

Remote, Spain

- Integrated an on-device AI inference engine using TensorFlowJS to process real-time audio streams. Trained and fine-tuned custom classification models to detect specific sleep patterns with high accuracy.
- Optimized the application architecture to handle heavy Machine Learning workloads directly on the mobile side (Edge AI), balancing high-performance inference with efficient resource utilization.
- Transformed raw audio telemetry into actionable data insights for parents, enhancing the core value proposition through reliable AI-driven features.

**Technical Lead & Full-Stack Developer at T2M Holding LLP**

**Oct 2017 — Apr 2019**

Remote, Colombia

- Led the engineering of enterprise-grade Conversational AI agents for healthcare and real estate sectors. Integrated Microsoft Bot Framework and Azure LUIS for precise intent recognition and entity extraction, establishing robust NLP workflows.
- Designed and maintained a distributed container orchestration system using Docker Swarm. Implemented CI/CD pipelines (GitLab) to balance rapid feature delivery with high code quality and system stability.
- Architected the data persistence layer using Couchbase (NoSQL) and MapReduce to process high volumes of interaction logs, enabling real-time analytics and business insights.

**Education**

**UNIMINUTO**

Tecnólogo en Informatica

**Bogota, Colombia**  
**Jan 2010 — Jan 2015**

**Certifications**

**Google Developer Expert**

**Jul 2019 — Present**

Angular & Web — recognized by Google for deep web expertise; I speak at conferences, publish technical content, mentor engineers, and contribute code samples to the community.

**Microsoft Most Valuable Professional**

**Apr 2022 — Present**

Developer Technologies — Microsoft-recognized community leadership through technical talks, content, and hands-on contributions to developer ecosystems.

**Platzi Instructor**

**Jan 2023 — Present**

Advanced curriculum for 30,000+ learners on Latin America's largest tech education platform, with emphasis on TypeScript, React, full-stack web (Angular, NestJS, Node.js), and applied AI for production systems. I translate complex architecture—scalable backends, LLM workflows, and reliability patterns—into practical courses developers can ship with.

**Skills**

**Programming Languages:** TypeScript, JavaScript, Python, SQL

**Frontend:** React, Next.js, Angular, Ionic, Tailwind CSS, Web Components, WebAssembly

**Backend and Infrastructure:** Node.js, NestJS, FastAPI, Serverless, Cloudflare Workers, Docker, PostgreSQL, Redis, NoSQL, GitHub Actions

**AI and ML:** LangChain, LangGraph, RAG, Prompt Engineering, TensorFlowJS, Voice AI, Google ADK

**Languages**

Spanish (native), English (B2)